Visualizing Normal Distributions Transform (NDT) Steps (TikZ Series)

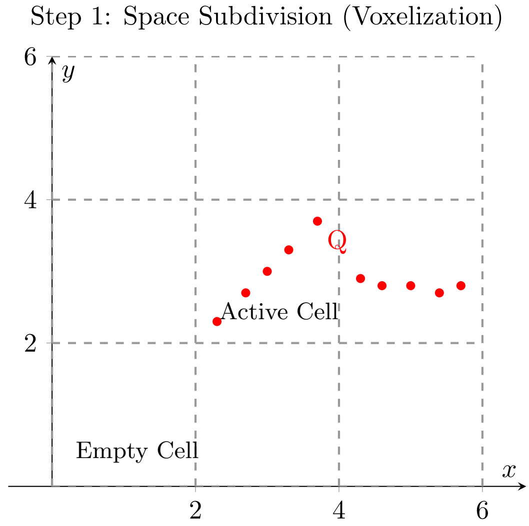

Figure 1: Step 1: The target point cloud (Q) is placed into a regular spatial grid (voxels in 3D, squares in 2D). Cells with fewer than a minimum number of points (e.g., 3 points) are usually ignored to prevent mathematical instability.

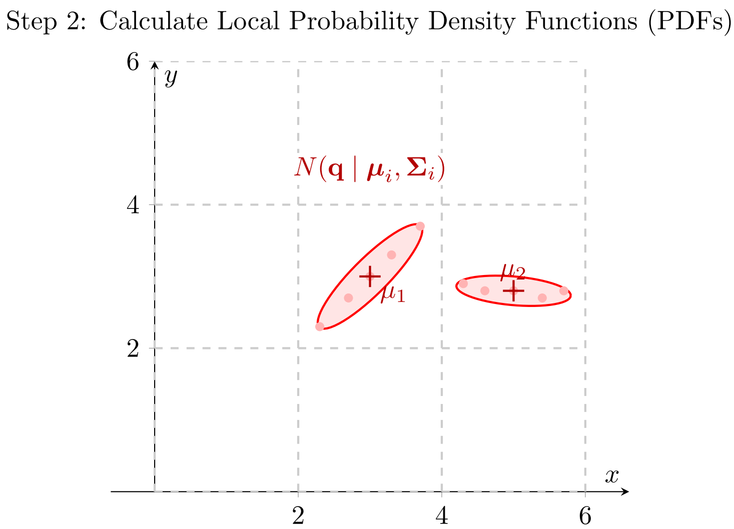

Figure 2: Step 2: For every active cell, the sample mean ($\boldsymbol{\mu}_i$) and covariance matrix ($\boldsymbol{\Sigma}_i$) of the target points are calculated. This transforms the discrete point cloud into a piecewise continuous set of multi-variate Gaussian probability distributions (represented here by covariance ellipses).

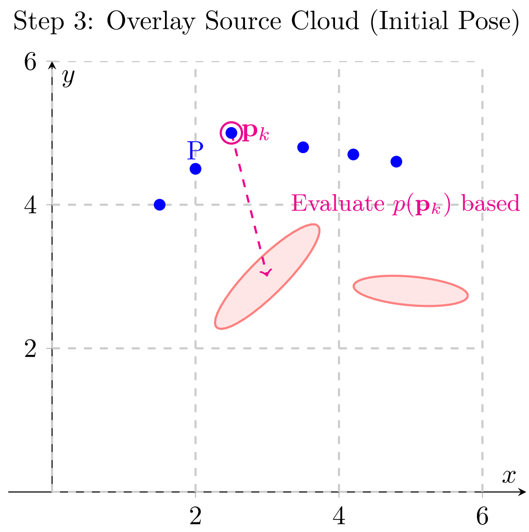

Figure 3: Step 3: The source point cloud (P) is projected into the voxel grid using its initial, unoptimized transformation. We determine which spatial cell each source point falls into (or its nearest neighbors if using smoothed NDT).

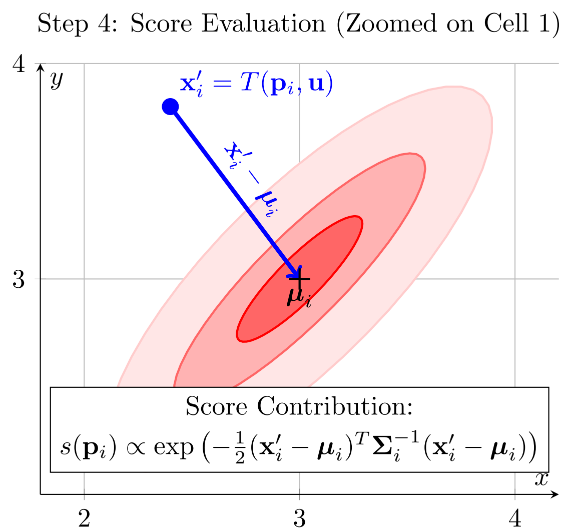

Figure 4: Step 4: NDT entirely drops discrete pairs. Instead, it evaluates the probability density of the target cell at the location of the transformed source point $\mathbf{x}'_i$. The objective function is to maximize the sum of these probabilities for all source points across the grid.

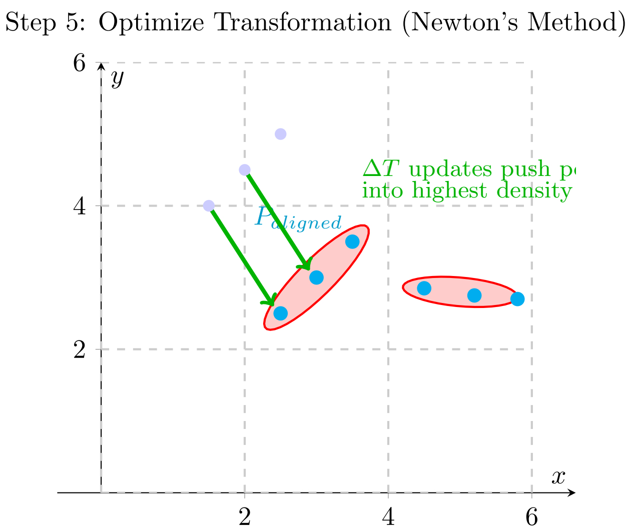

Figure 5: Step 5: An optimization algorithm (typically Newton's Method) calculates the Hessian and gradient of the objective function to update the transformation parameters (rotation and translation). The source cloud "falls" into the high-probability valleys defined by the target's Gaussian mixture.